

Regression with a categorical predictor

Reminders and agenda

- + Exam 1 next Friday (Feb 27)
 - + Covers material through the last class (Feb 16)
- + Exam 1 review day next Wednesday (Feb 25)
- + Studying advice:
 - + Practice questions on the course website
 - + Review previous class activities
 - + Review previous homework assignments
- + Today: thinking about categorical explanatory variables

Regression so far

Simple linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

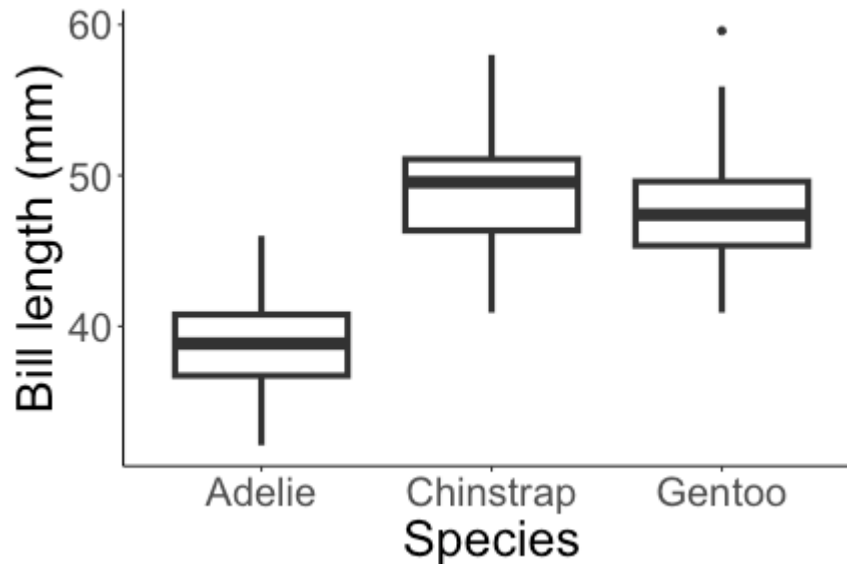
Polynomial regression:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k + \varepsilon$$

where x and y are both **quantitative**.

What if x is *categorical*?

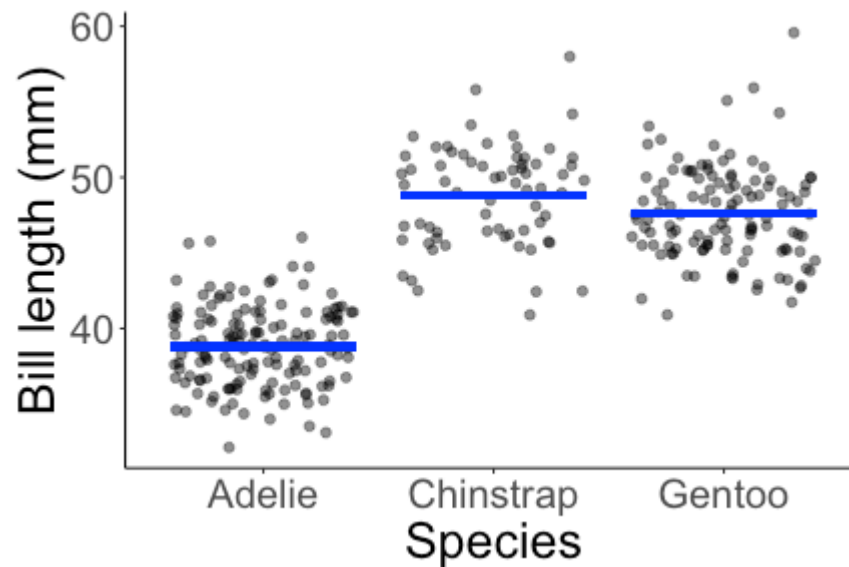
Categorical predictor



- + How do we model the relationship between species and bill length?
- + Can we test for a relationship between bill length and species?

Categorical predictor

Imagine population like this:

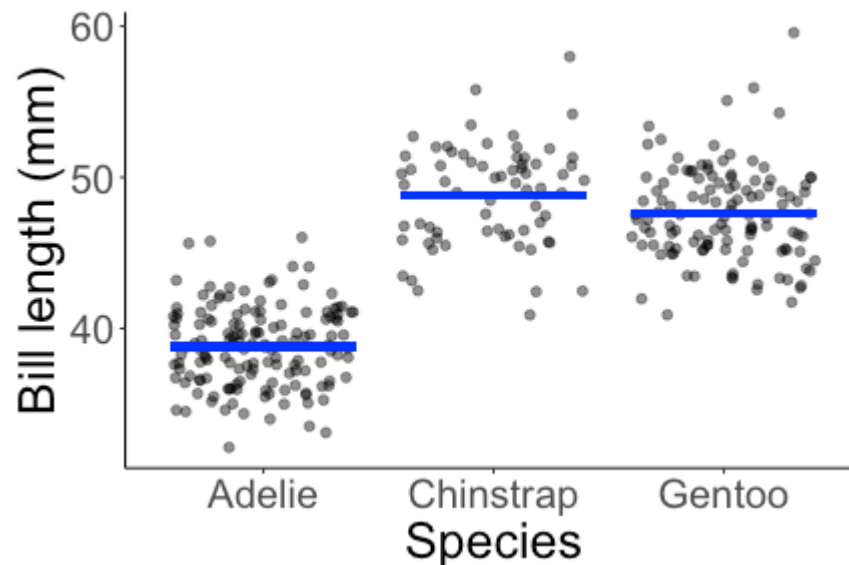


If species = Adelie,

$$\text{bill length} = \text{mean Adelie bill length} + \varepsilon$$

Categorical predictor

Imagine population like this:

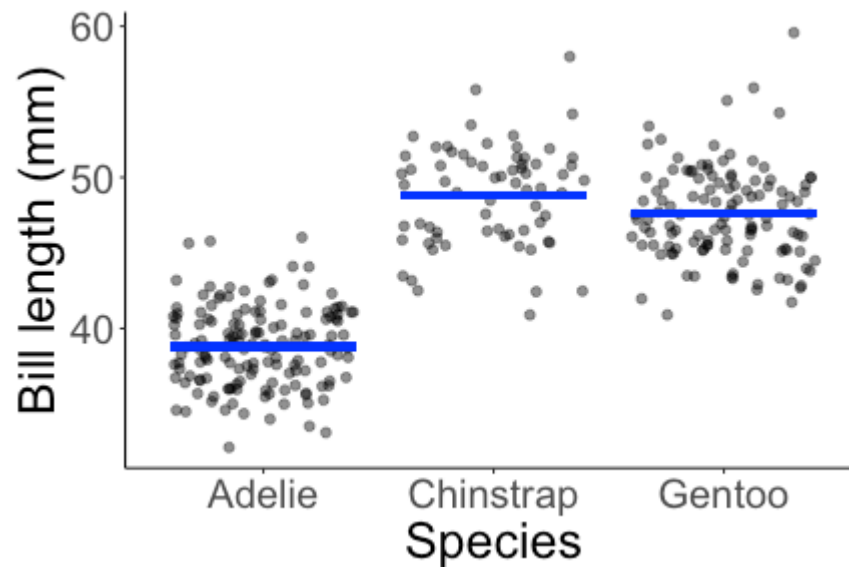


If species = Chinstrap,

$$\text{bill length} = \text{mean Chinstrap bill length} + \varepsilon$$

Categorical predictor

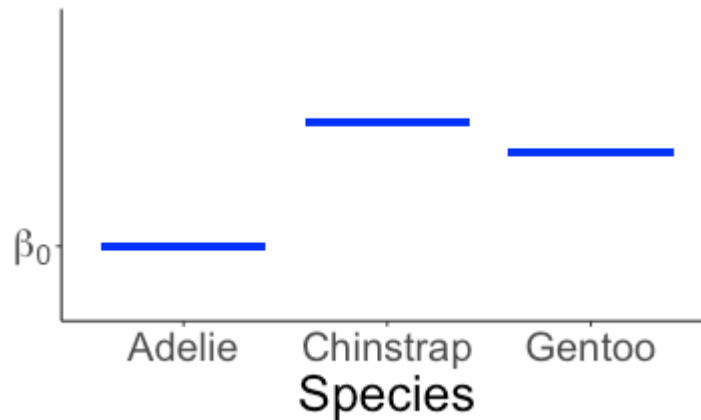
Imagine population like this:



If species = Gentoo,

$$\text{bill length} = \text{mean Gentoo bill length} + \varepsilon$$

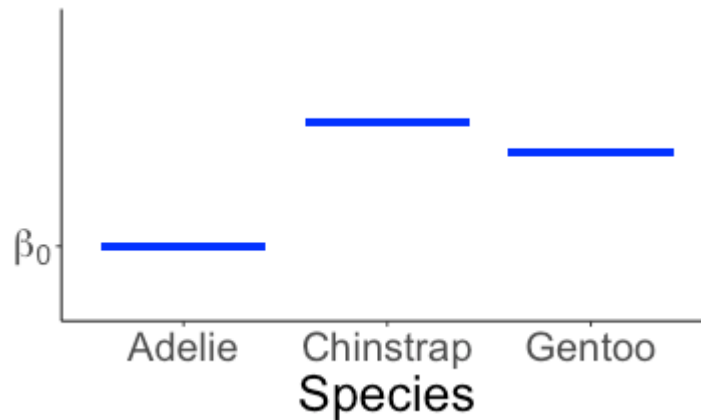
Categorical predictor



If species = Adelie, bill length = $\beta_0 + \varepsilon$

- + β_0 = true mean bill length of Adelie penguins (in the population)

Categorical predictor

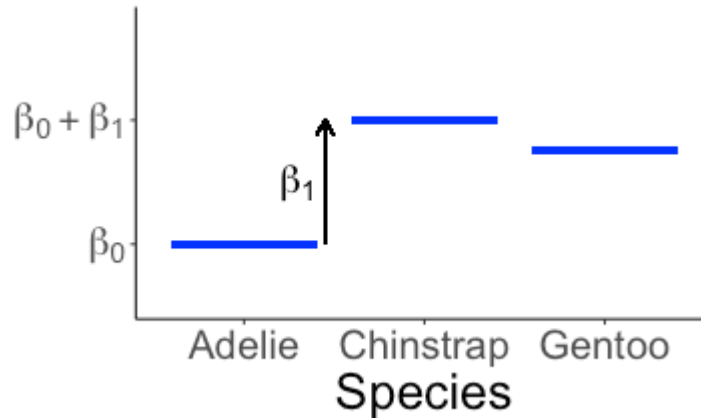


If species = Adelie, bill length = $\beta_0 + \varepsilon$

+ β_0 = true mean bill length of Adelie penguins (in the population)

What notation could I use for Chinstrap penguins?

Categorical predictor

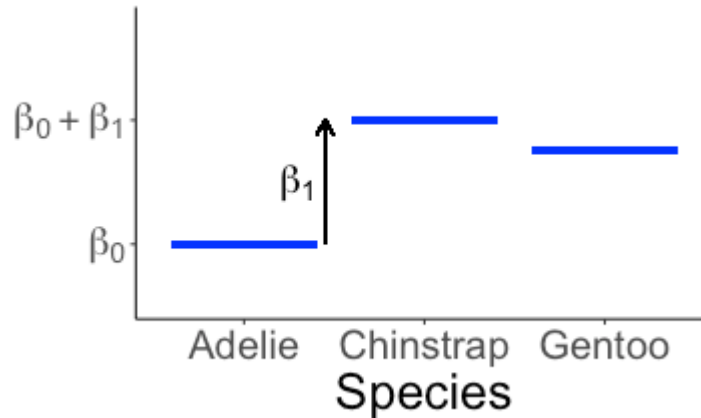


If species = Adelie, bill length = $\beta_0 + \varepsilon$

+ β_0 = true mean bill length of Adelie penguins (in the population)

If species = Chinstrap, bill length = $\beta_0 + \beta_1 + \varepsilon$

Categorical predictor

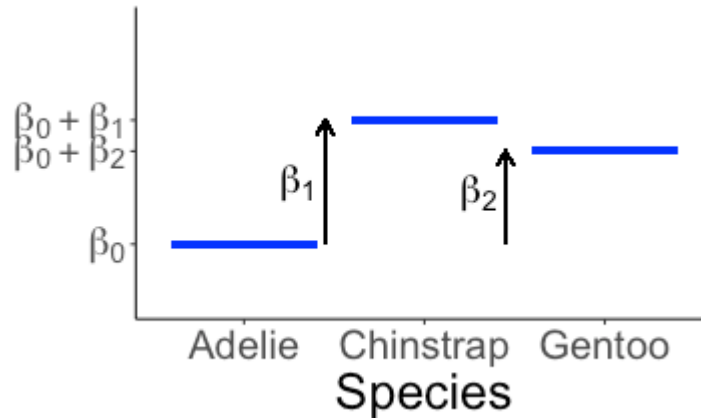


If species = Adelie, bill length = $\beta_0 + \varepsilon$

If species = Chinstrap, bill length = $\beta_0 + \beta_1 + \varepsilon$

What about Gentoo penguins?

Categorical predictor

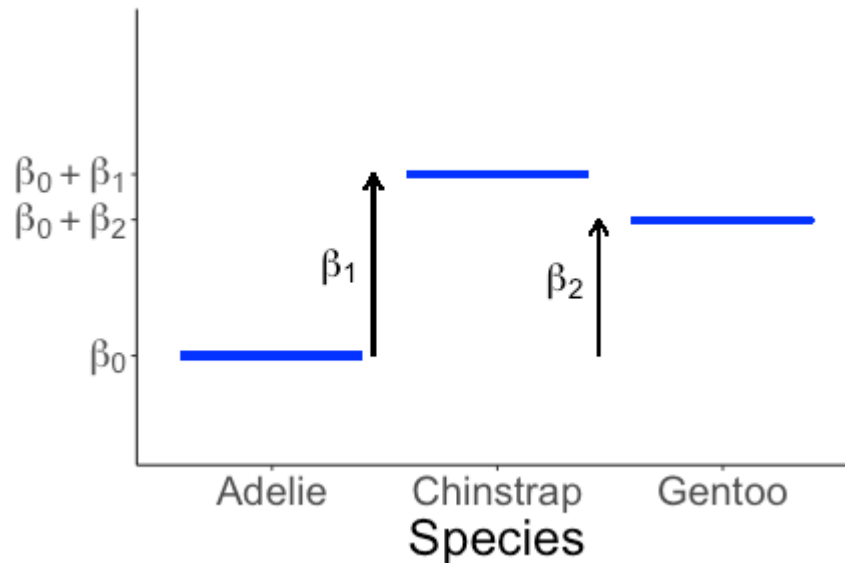


If species = Adelie, bill length = $\beta_0 + \varepsilon$

If species = Chinstrap, bill length = $\beta_0 + \beta_1 + \varepsilon$

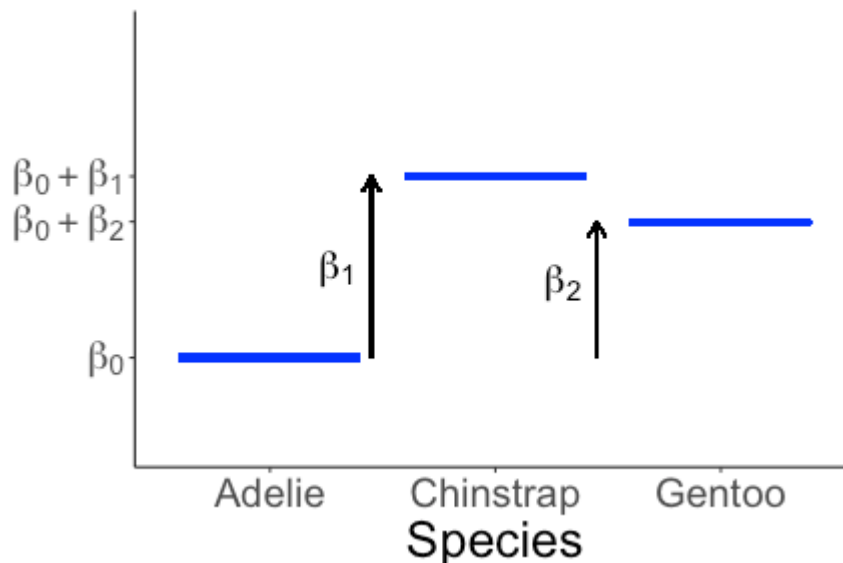
If species = Gentoo, bill length = $\beta_0 + \beta_2 + \varepsilon$

Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Can we write this more concisely?

Indicator variables

$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Do I need another indicator variable to represent Adelie penguins?

Indicator variables

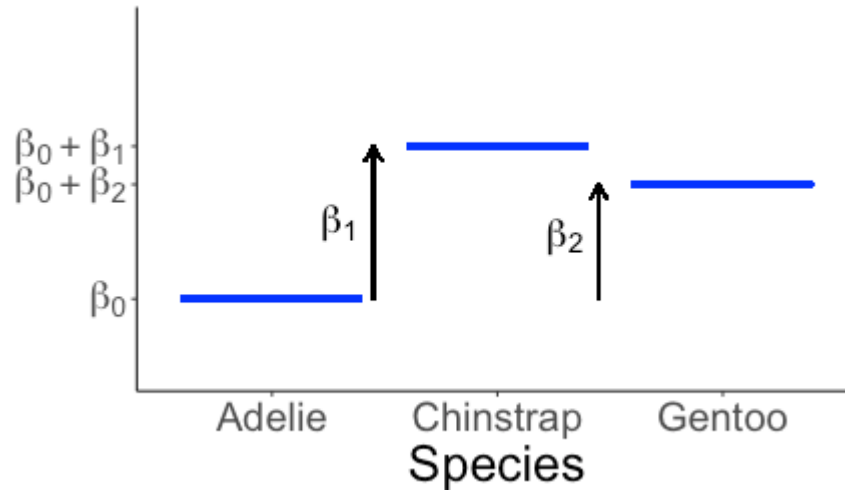
$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Then:

Species	IsChinstrap	IsGentoo
Adelie	0	0
Chinstrap	1	0
Gentoo	0	1

Indicator variables



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

- + What is the estimated average bill length for each of the three species of penguin?
- + There are 333 penguins in the dataset used to fit the model. How many degrees of freedom does the fitted model have?

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

Species	$\widehat{\text{bill length}}$
Adelie	38.82
Chinstrap	$38.82 + 10.01 = 48.83$
Gentoo	$38.82 + 8.74 = 47.56$

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

Species	$\widehat{\text{bill length}}$
Adelie	38.82
Chinstrap	$38.82 + 10.01 = 48.83$
Gentoo	$38.82 + 8.74 = 47.56$

Degrees of freedom: $333 - 3 = 330$

+ (Lose one degree of freedom for each parameter estimated)

Fitting the model in R

R will automatically create indicator variables:

```
lm(bill_length_mm ~ species, data = penguins_no_nas)
```

```
##
```

```
## Call:
```

```
## lm(formula = bill_length_mm ~ species, data = penguins_n
```

```
##
```

```
## Coefficients:
```

```
##          (Intercept)    speciesChinstrap    speciesGentoo
```

```
##          38.824          10.010          8.744
```

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

Class activity

- + Practice regression with a single categorical predictor, with experimental data on fruit flies

https://sta112-s26.github.io/class_activities/ca_15.html